atomic mass unit (*a.m.u.*) which is approximately equal to the mass of a hydrogen atom, or of a neutron; it is exactly $\frac{1}{16}$ of the mass of the oxygen isotope ${}_8O^{16}$. The mass defects are of the order of magnitude of 1/1,000 a.m.u., and therefore it is customary to express them in units of 1/1,000 a.m.u. (a.m.u.$^{-3}$). The energy equivalent of 1 a.m.u.$^{-3}$, i.e. 1 a.m.u.$^{-3} \times c^2$, happens to differ only slightly from the energy unit 1 Mev. To be exact:

1 a.m.u.$^{-3}$ is the equivalent of 0.93 Mev.

Accordingly, the binding energies of nuclei are also of the order of magnitude of 1 Mev.

Thus the physicist has two completely independent processes available for determining the binding energies: He can measure them directly, or can compute them from the mass defects. In the latter case, it is essential, of course, to be able to determine the masses of the atoms to a high degree of accuracy, in order to deduce the binding energy from them, for we are dealing with minute differences, measurable in 1/1,000ths of the masses only. Let it suffice here to say that this is accomplished with the aid of the *mass spectrograph,* an apparatus first developed by Aston. In the mass spectrograph, charged atoms are caused to pass through electric and magnetic fields, where they are deflected. As we have already seen, the extent of their deflections depends on the ratio of their charge *e* to their mass *m*, as well as on their velocity. The mass spectrograph is designed so that only particles having a certain velocity are singled out and become the objects of observation. The ratio *e/m,* can then be determined, and since the charge *e* of the particles is known, their mass *m* can be computed—or, to be more exact, their mass, *m,* can be compared with the mass of the oxygen isotope ${}_8O^{16}$, which is exactly 16.000 a.m.u. by definition. For the calculation of the binding energies, we need also the exact masses of the hydrogen atom and the neutron. The mass of the hydrogen atom is 1.00813 a.m.u., the mass of the neutron is 1.00895 a.m.u.

Let us now consider, quantitatively, the changes in mass when a deuterium atom is formed. Before the formation of this atom, there was an individual hydrogen atom, ${}_1H^1$, and an individual neutron, ${}_0n^1$; after the combination of these two, there is a deuterium atom, ${}_1D^2$, and a free photon, *hv.* This finding can be expressed by the following formula: